Module 1 — Development of Practical Skills in Chemistry

1.1 Planning Experiments

Hypothesis: A testable prediction based on scientific knowledge or prior observations that proposes a relationship between variables. A good hypothesis is specific and falsifiable — for example, "Increasing the concentration of hydrochloric acid will increase the rate of reaction with magnesium ribbon, measured by the volume of hydrogen gas produced in 60 seconds."

Variables

Every experiment involves three types of variables that must be clearly identified during the planning stage:

- **Independent variable:** The variable you deliberately change (e.g. concentration of acid). You should plan at least five different values across a suitable range to identify a clear trend.
- **Dependent variable:** The variable you measure as a result (e.g. volume of gas produced). Choose a method of measurement that gives quantitative, continuous data where possible.
- **Control variables:** All other variables that must be kept constant to ensure a fair test (e.g. temperature, volume of reactant, catalyst presence, surface area of solid). Failing to control these introduces confounding variables that undermine validity.

Exam Tip: When planning experiments, always explain *how* you will control each variable, not just state that you will. For example: "Temperature will be controlled by using a water bath set to 25 °C, monitored with a thermometer."

Risk Assessments

A risk assessment identifies hazards associated with the chemicals and procedures, evaluates the likelihood and severity of harm, and outlines precautions to minimise risk. You should reference standard hazard symbols (GHS pictograms) and safety data sheets (SDS). In an exam, structure your answer as: **Hazard → Risk → Precaution**.

Hazard Symbol	Meaning	Example Chemical	Precaution
Corrosive	Destroys living tissue on contact	Concentrated sulfuric acid (H_2SO_4)	Wear gloves, goggles; use in fume cupboard; neutralise spills with sodium hydrogencarbonate
Toxic	Can cause death or serious harm if inhaled, swallowed, or absorbed	Chlorine gas (Cl_2)	Use fume cupboard; small quantities only; know location of emergency eyewash
Flammable	Catches fire easily; flash point below 60 °C	Ethanol (C_2H_5OH)	Keep away from naked flames; heat with water bath or electric heater, never a Bunsen burner
Irritant / Harmful	Causes redness, inflammation, or mild harm	Dilute hydrochloric acid (HCl)	Wear goggles; wash skin immediately if splashed
Oxidising	Provides oxygen; can intensify fires and cause explosions with flammable materials	Potassium manganate(VII) ($KMnO_4$)	Keep away from flammable materials; store separately
Environmental hazard	Toxic to aquatic organisms	Copper sulfate ($CuSO_4$)	Do not pour down the drain; dispose of in designated waste containers

Selecting Apparatus

You must choose apparatus appropriate to the measurement required. Consider the **resolution** (smallest scale division), the **range** (maximum value it can measure), and the **uncertainty** of the instrument. Always select the instrument that gives the lowest percentage uncertainty for the quantity being measured.

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Apparatus	Resolution	Typical Uncertainty	Best Use
Burette (50 cm^3)	0.10 cm^3	±0.05 cm^3 per reading	Delivering variable, measured volumes in titrations
Graduated pipette (25.0 cm^3)	0.10 cm^3	±0.06 cm^3	Transferring a fixed, precise volume
Measuring cylinder (100 cm^3)	1 cm^3	±0.5 cm^3	Measuring approximate volumes
Volumetric flask (250 cm^3)	Single graduation	±0.20 cm^3	Making standard solutions of known concentration
Balance (2 d.p.)	0.01 g	±0.01 g	Measuring masses for moles calculations
Balance (3 d.p.)	0.001 g	±0.001 g	Precise mass measurements for gravimetric analysis
Thermometer (−10 to 110 °C)	0.5 °C or 1 °C	±0.5 °C	Measuring temperature changes in calorimetry
Gas syringe (100 cm^3)	1 cm^3	±0.5 cm^3	Collecting and measuring volumes of gas

Exam Tip: When asked to describe an experimental procedure, always state the specific apparatus you would use and why. For example, state "use a burette (resolution ±0.05 cm^3)" rather than simply "measure the volume." The key phrase is "appropriate to the scale of measurement."

Making Standard Solutions

A **standard solution** is one of accurately known concentration, prepared using a volumetric flask:

1. Calculate the mass of solute required using $n = c \times V$ and $m = n \times M$.
2. Weigh the solute accurately on a balance (record to ±0.01 g or better).

3. Dissolve the solute in a small volume of distilled water in a beaker, stirring with a glass rod.
4. Transfer the solution to a volumetric flask using a funnel. Rinse the beaker and glass rod with distilled water and add the washings to the flask to ensure all solute is transferred.
5. Add distilled water to the flask up to the calibration mark, using a dropping pipette for the final drops. The bottom of the meniscus must sit exactly on the line.
6. Stopper and invert the flask several times to ensure thorough mixing.

1.2 Recording and Presenting Results

Recording Data

Raw data should be recorded in a clearly labelled table with appropriate headings that include units. Headings should follow the format: *Quantity / Unit* (e.g. "Volume / cm^3"). All readings should be recorded to the same number of decimal places, consistent with the precision of the instrument. Never round raw data during recording.

Qualitative observations should also be recorded systematically. For colour changes, describe both the initial and final colour (e.g. "orange solution turned green," not just "colour changed").

Significant Figures

Calculated results should be given to the same number of significant figures as the least precise measurement used. For example, if a mass is measured to 3 significant figures and a volume to 4, the final answer should be reported to 3 significant figures.

Significant figures: The number of meaningful digits in a measurement, starting from the first non-zero digit. For example, 0.00452 has 3 significant figures. Trailing zeros after a decimal point are significant (e.g. 2.50 has 3 s.f.), but leading zeros are not.

Exam Tip: A common mistake is giving too many significant figures in a final answer. If your raw data is to 3 s.f., your answer cannot be more precise than 3 s.f. However, keep extra figures during intermediate calculations and only round at the very end to avoid rounding errors accumulating.

Graphs

Graphs should have:

- Labelled axes with units (in the format "Quantity / unit")
- An appropriate scale that uses at least half the available graph paper in both directions
- Data points plotted accurately as small crosses (×), not dots or circles
- A line of best fit (straight or curved as appropriate), which does not need to pass through every point but should have roughly equal numbers of points above and below
- A descriptive title
- Anomalous points clearly identified (circled) and excluded from the line of best fit

When calculating a gradient, use a large triangle (at least half the length of the line) and show the construction on the graph. Read coordinates from the line of best fit, not from data points.

1.3 Analysis and Evaluation

Accuracy, Precision, Reliability and Validity

Accuracy: How close a measured value is to the true or accepted value. Affected by systematic errors.

Precision: How close repeated measurements are to each other, regardless of whether they are close to the true value. A precise set of results has a small spread (low standard deviation). Affected by random errors.

Reliability (Repeatability): Results are reliable if they can be consistently reproduced under the same conditions by the same experimenter. **Reproducibility** is achieved when different experimenters using different equipment obtain similar results.

Validity: An experiment is valid if it measures what it claims to measure. This requires that all control variables are properly managed and the method is appropriate for the investigation.

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Exam Tip: Learn the distinction between these four terms precisely. A common exam error is confusing accuracy with precision. An analogy: if you throw five darts and they all land clustered tightly together but far from the bullseye, your throws are *precise but not accurate*. If they are scattered around the bullseye, they are *accurate (on average) but not precise*.

Types of Error

Error Type	Description	Effect	How to Reduce
Random error	Unpredictable fluctuations in readings caused by uncontrollable factors (e.g. slight variations in reading a meniscus, timing reflexes)	Reduces precision; scatter around the true value	Take repeat measurements and calculate a mean; use data-logging equipment; plot a graph and draw a line of best fit
Systematic error	Consistent error in the same direction every time (e.g. a balance that always reads 0.02 g too high, heat loss in calorimetry)	Reduces accuracy; shifts all readings in the same direction	Calibrate instruments before use; improve technique; use more insulated apparatus
Zero error	Instrument does not read zero when it should (a type of systematic error)	Systematic shift of all readings by a fixed amount	Check and reset the instrument before use; subtract the zero error from all readings
Parallax error	Reading a scale from an angle rather than at eye level	Systematic or random depending on consistency	Always read at eye level; for burettes, read the bottom of the meniscus

Percentage Error and Uncertainty

$$\text{Percentage error} = (|\text{experimental value} - \text{true value}| \,/\, \text{true value}) \times 100\%$$

$$\text{Percentage uncertainty} = (\text{absolute uncertainty} \,/\, \text{measured value}) \times 100\%$$

For a burette with an uncertainty of ±0.05 cm^3 per reading, two readings are taken (initial and final), so the total uncertainty is ±0.10 cm^3. If the titre is 25.00 cm^3, the percentage uncertainty is (0.10 / 25.00) × 100 = 0.40%.

Worked Example — Combining Uncertainties:

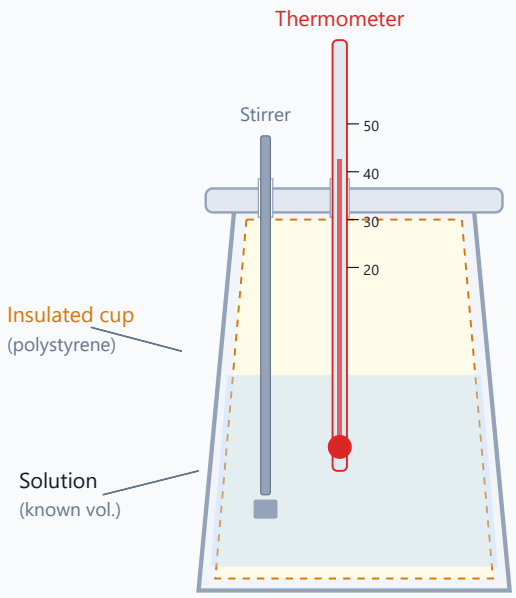
A student determines the concentration of NaOH by titration. The burette has an uncertainty of ±0.05 cm^3 per reading. The mean titre is 23.40 cm^3. The pipette delivers 25.0 cm^3 with an uncertainty of ±0.06 cm^3.

Burette uncertainty = 2 × 0.05 = ±0.10 cm^3 → % uncertainty = (0.10 / 23.40) × 100 = 0.43%
Pipette uncertainty = ±0.06 cm^3 → % uncertainty = (0.06 / 25.0) × 100 = 0.24%

Total % uncertainty = 0.43 + 0.24 = **0.67%**

The burette contributes the most uncertainty, so to improve the experiment, you could use a larger titre volume (reducing the percentage uncertainty of the burette reading) or use a more precise burette.

Calorimetry Setup (Measuring Enthalpy Changes)



Key Equation

$$q = mc\Delta T$$

m = mass of solution (g)
c = 4.18 J g^{-1} K^{-1} (for water)
ΔT = temperature change (K)

Sources of Error

- Heat loss to surroundings
- Incomplete combustion
- Heat absorbed by cup/therm.
- Evaporation of solution
- Density assumed = 1.00 g/cm^3

Improve: use lid + insulation

Exam Tip: When combining uncertainties, **add absolute uncertainties** for addition/subtraction and **add percentage uncertainties** for multiplication/division. The largest percentage uncertainty identifies the piece of apparatus that limits the overall precision — suggest improving this apparatus first.

Evaluating Procedures

In evaluation questions, you should:

1. Identify the most significant sources of error (not trivial ones like "I might have spilt some").
2. Explain the direction of error — does it make the result too high or too low?
3. Suggest specific, practical improvements to reduce those errors.
4. Assess whether the results support the hypothesis with reference to the data.
5. Consider whether anomalous results should be excluded from mean calculations and explain why they occurred.
6. Calculate the percentage error compared to the accepted value and comment on accuracy.

1.4 Practical Techniques

Titrations

Titration: A quantitative technique used to determine the concentration of a solution by reacting it with a solution of known concentration until the equivalence point is reached. The equivalence point is the exact point where the moles of reactants are in the stoichiometric ratio.

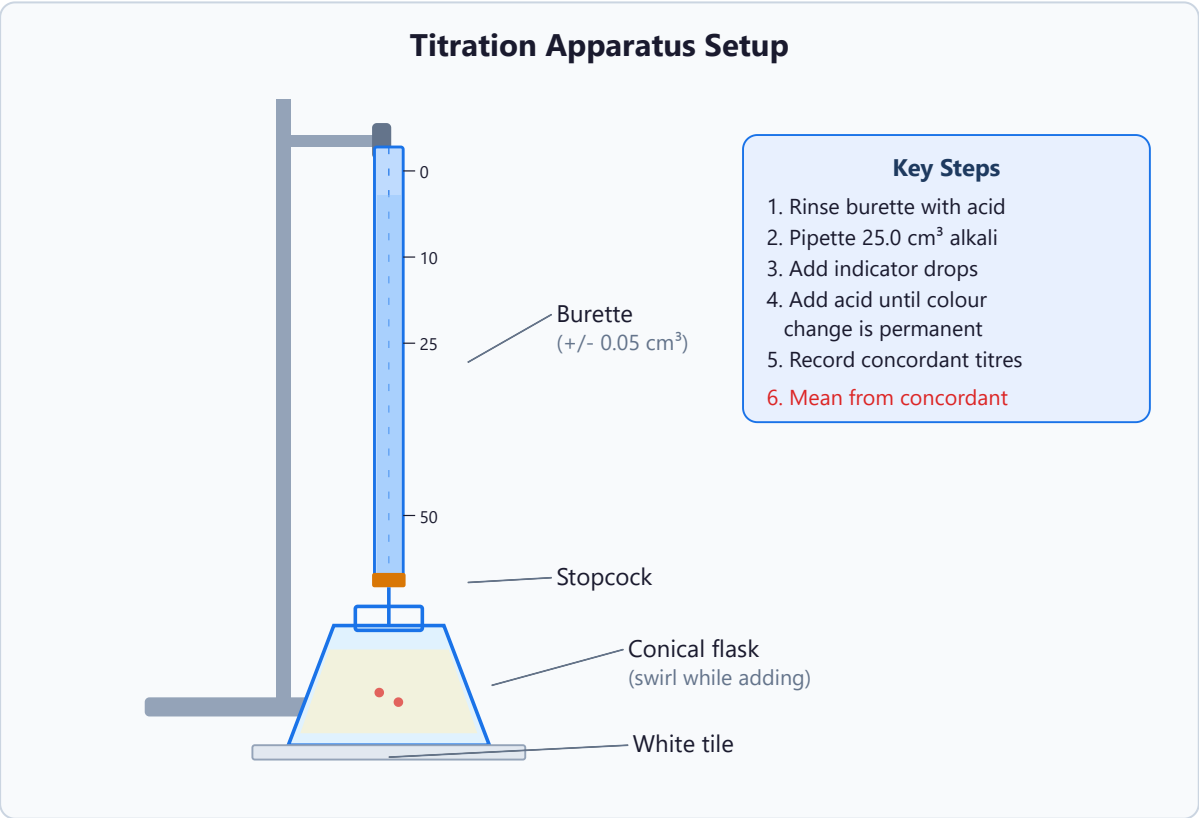
Standard procedure for an acid-base titration:

1. Rinse the burette with the solution it will contain (e.g. the acid) to prevent dilution. Fill and record the initial reading to 2 decimal places (e.g. 0.50 cm^3, not 0.5).
2. Use a pipette (rinsed with the solution it will contain) to transfer a fixed volume (e.g. 25.0 cm^3) of the other solution into a conical flask. The pipette should be allowed to drain freely — do not blow out the last drop.
3. Add 2–3 drops of a suitable indicator:

- **Methyl orange:** for strong acid + weak base (red in acid, yellow in alkali; endpoint is orange)
- **Phenolphthalein:** for weak acid + strong base (colourless in acid, pink in alkali)
- For strong acid + strong base, either indicator works as the pH change at equivalence is very sharp

4. Add the solution from the burette slowly, swirling the flask constantly for mixing, until the indicator changes colour permanently (lasting at least 30 seconds). Record the final burette reading.
5. Perform a rough titration first, then repeat to obtain at least two concordant results (within 0.10 cm^3 of each other).
6. Calculate the mean titre from concordant results only, discarding the rough and any anomalous values.

Exam Tip: The conical flask is rinsed with deionised water only (not with the solution), because adding extra solution would change the amount of substance present and alter the result. The pipette and burette are rinsed with their respective solutions to avoid dilution. A white tile is placed underneath the flask to help observe the colour change more clearly. These are common 1-mark points in practical questions.



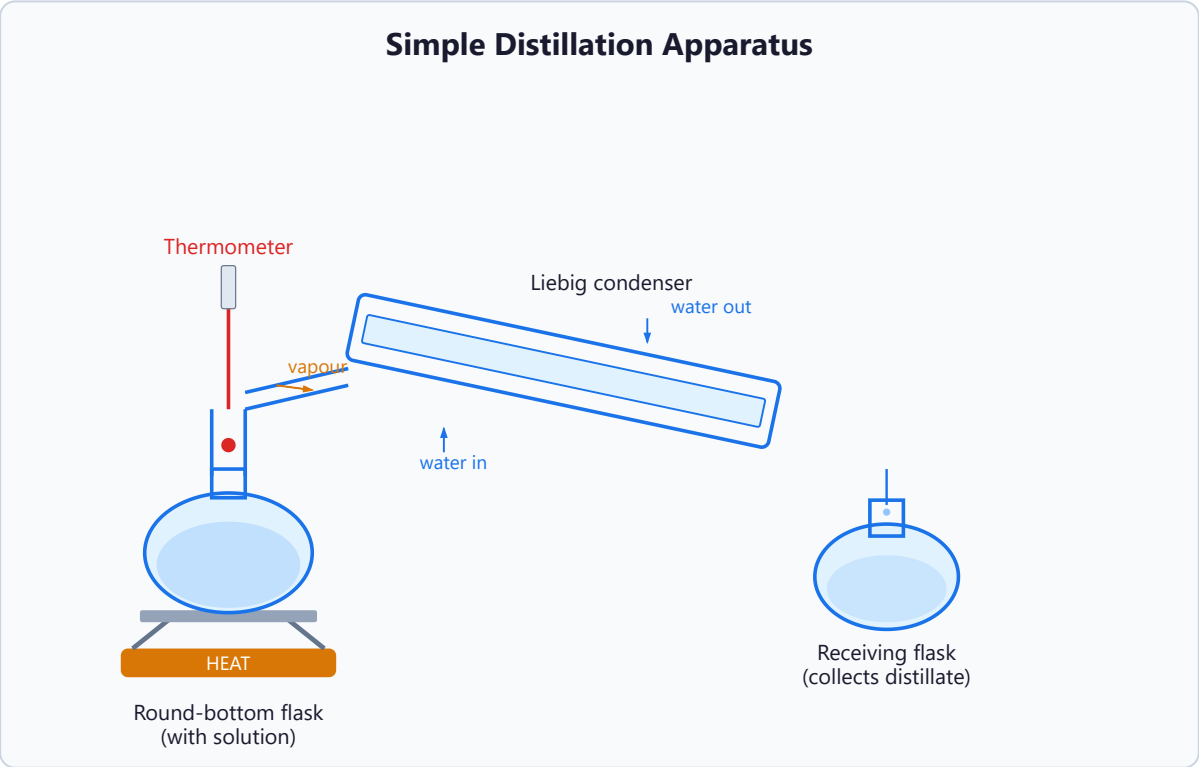
Worked Example — Titration Calculation:

25.0 cm^3 of 0.100 mol dm^3 NaOH was titrated against HCl of unknown concentration. The mean concordant titre was 22.50 cm^3.

Step 1: Moles of NaOH = c × V = 0.100 × (25.0/1000) = 2.50×10^3 mol
Step 2: From the equation NaOH + HCl → NaCl + H_2O, the ratio is 1:1, so moles of HCl = 2.50×10^3 mol
Step 3: Concentration of HCl = n / V = 2.50×10^3 / (22.50/1000) = **0.111 mol dm^3** (3 s.f.)

Distillation

Simple distillation separates a solvent from a solution (e.g. water from salt water) or separates liquids with boiling points differing by more than about 25 °C. The solution is heated, the solvent evaporates, passes through a Liebig condenser where it cools and condenses, and is collected in a receiving flask. The thermometer at the top monitors the vapour temperature to ensure you are collecting the correct fraction.

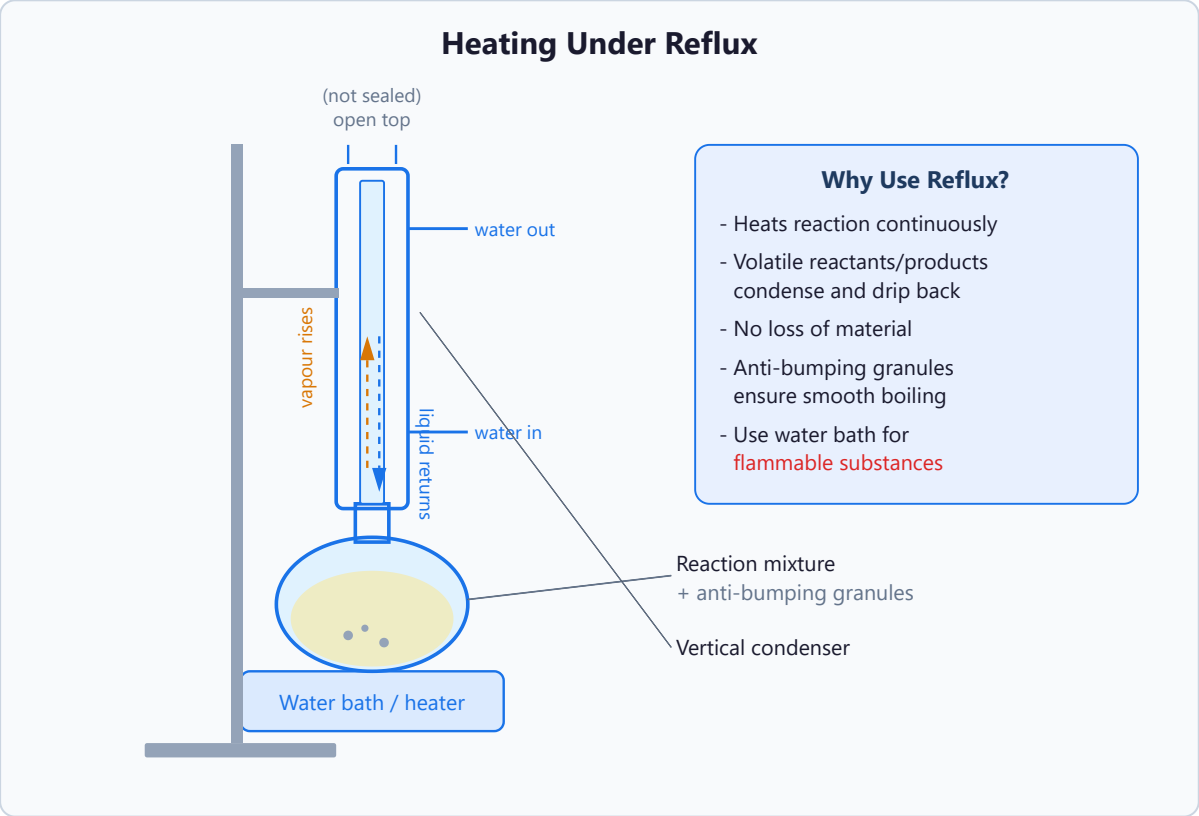


Fractional distillation separates a mixture of miscible liquids with similar boiling points (within about 25 °C of each other), such as ethanol and water. A fractionating column is packed with glass beads or has

indentations to provide a large surface area for repeated vaporisation-condensation cycles. The liquid with the lowest boiling point reaches the top of the column first and is collected. A thermometer at the top monitors which fraction is being collected.

Reflux

Reflux: A technique for heating a reaction mixture continuously over an extended period without losing volatile components. Vapour rises, condenses in a vertical condenser attached to the flask, and drips back into the mixture. The apparatus is **never sealed** (open at the top of the condenser) to prevent dangerous pressure build-up.



Reflux is used when a reaction requires prolonged heating (e.g. oxidation of a primary alcohol with acidified potassium dichromate to form a carboxylic acid, or esterification). An anti-bumping granule (porous ceramic chip) is added to ensure smooth boiling by providing nucleation sites for bubble formation. The flask is heated using a water bath or electric heater rather than a naked flame when flammable substances are involved.

Key distinction: Reflux is used to *carry out a reaction* that needs sustained heating. Distillation is used to *separate and collect* a product or component. Sometimes both are used in sequence: reflux first (to

complete the reaction), then distillation (to collect the volatile product).

Recrystallisation

Recrystallisation purifies an impure solid by exploiting differences in solubility at different temperatures. The underlying principle is that most solids are more soluble in hot solvents than cold solvents:

1. **Dissolve** the impure solid in the minimum volume of hot solvent. Using the minimum volume is critical — too much solvent means less product crystallises out, reducing yield.
2. **Hot filter** the solution to remove any insoluble impurities (this is called hot filtration using a pre-heated funnel and fluted filter paper to prevent premature crystallisation).
3. **Cool slowly.** Allow the filtrate to cool slowly to room temperature, then further in an ice bath. The desired product crystallises out as it becomes less soluble at lower temperatures. Impurities remain dissolved because they are present in small amounts below their saturation point.
4. **Filter** the crystals using vacuum (Buchner) filtration for faster and drier collection.
5. **Wash** the crystals with a small volume of ice-cold solvent to remove residual impurities from the crystal surfaces. Use cold solvent to minimise re-dissolving your product.
6. **Dry** the crystals between filter papers, in an oven at low temperature, or in a desiccator.

Assessing purity: Measure the melting point of your purified product. A pure substance has a sharp melting point that matches the literature value. Impurities lower the melting point and broaden the melting range.

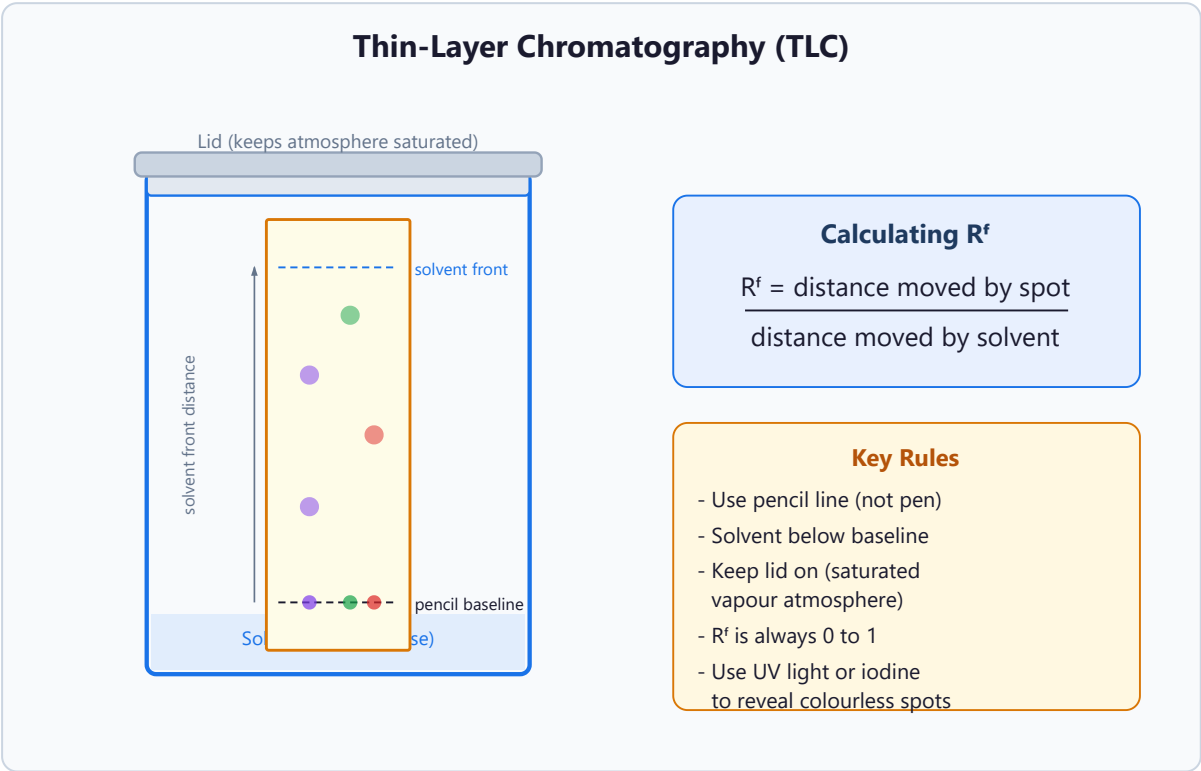
Chromatography

Chromatography separates components of a mixture based on their different affinities for a **stationary phase** (the fixed medium) and a **mobile phase** (the moving solvent or gas). Components with a greater affinity for the mobile phase travel faster and further.

Thin-layer chromatography (TLC): A small spot of sample is placed on a TLC plate (stationary phase: thin layer of silica or alumina on a glass or plastic plate). The plate is placed in a beaker with a shallow layer of solvent (mobile phase). The solvent rises by capillary action, carrying components at different rates. An R_f value is calculated for each spot:

$$R_f = \text{distance moved by spot} / \text{distance moved by solvent front}$$

Paper chromatography works on the same principle but uses chromatography paper as the stationary phase (water trapped in the cellulose fibres acts as the true stationary phase). Both techniques can be used to identify unknown substances by comparing R_f values with those of known standards run on the same plate or paper under identical conditions.



Exam Tip: The pencil baseline must be used (not pen) because ink would dissolve in the solvent and interfere with results. The solvent level must be below the baseline so spots are not dissolved directly into the solvent. The lid must be on the beaker so the atmosphere is saturated with solvent vapour, preventing uneven evaporation from the plate. These are all common exam marks.

Gravimetric Analysis

Gravimetric analysis: A quantitative technique that determines the amount of a substance by measuring mass changes, often involving precipitation, filtration, drying and weighing. The key principle is that a substance of interest is converted into an insoluble compound of known formula, which is then isolated and weighed.

A typical gravimetric procedure involves adding an excess of a reagent to precipitate the desired ion from solution, then filtering, washing, drying and weighing the precipitate. For example, determining the sulfate

content of a solution by adding excess barium chloride to precipitate barium sulfate:

$$Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$$

The mass of the dried precipitate is used to calculate the moles and hence the mass or concentration of the sulfate ions originally present. Drying to **constant mass** means heating, cooling in a desiccator, and weighing repeatedly until successive weighings agree to within ±0.01 g.

Worked Example — Gravimetric Analysis:

A 50.0 cm^3 sample of solution containing sulfate ions was treated with excess $BaCl_2$. The precipitate of $BaSO_4$ was filtered, washed, dried and weighed. Mass = 0.466 g.

Step 1: M_r of $BaSO_4$ = 137.3 + 32.1 + (4 × 16.0) = 233.4

Step 2: Moles of $BaSO_4$ = 0.466 / 233.4 = 2.00×10^{-3} mol

Step 3: From the equation, moles of SO_4^{2-} = 2.00×10^{-3} mol (1:1 ratio)

Step 4: Concentration of SO_4^{2-} = 2.00×10^{-3} / 0.0500 = **0.0400 mol dm^{-3}**

Other Key Practical Techniques

Technique	Purpose	Key Points
Vacuum (Buchner) filtration	Faster filtration; drier solid product	Uses a Buchner funnel, filter flask, and vacuum pump or water aspirator. Wet the filter paper with solvent first to create a seal. More efficient than gravity filtration for collecting precipitates and recrystallised products.
Gravity filtration	Remove insoluble impurities from a solution	Uses a filter funnel and filter paper in a conical flask. Slower but simpler. Used for hot filtration during recrystallisation (use a fluted filter paper and a pre-heated funnel).
Water bath heating	Gentle, controlled heating of flammable liquids	Avoids naked flames; maintains more even temperature than direct heating. Essential when working with ethanol, propanone, and

		other flammable organic solvents. An electric heating mantle is an alternative.
Use of a desiccator	Drying solids to constant mass	Contains a drying agent (e.g. anhydrous calcium chloride, silica gel); weigh, dry, re-weigh until mass is constant (±0.01 g). Removes atmospheric moisture. Used in gravimetric analysis and determining water of crystallisation.
Measuring pH	Determine acidity or alkalinity	Use a calibrated pH meter for quantitative accuracy (resolution ±0.01 pH units) or universal indicator paper for a rough qualitative estimate. Calibrate the pH meter using standard buffer solutions before use.
Collecting gases	Measure gas volume produced in a reaction	Use a gas syringe for accurate measurement, or collect over water using an inverted measuring cylinder (only for insoluble gases). Ensure all joints are airtight to prevent gas loss.

1.5 Practical Endorsement (PAG Summary)

The OCR A specification includes a set of Practical Activity Groups (PAGs) that must be completed to gain the practical endorsement. These assess your ability to follow written instructions, use appropriate apparatus, make and record observations, and analyse results. You need to demonstrate competence in **five key areas**: following procedures, applying investigative approaches, using instruments and equipment, making and recording observations, and researching and referencing.

- **PAG 1:** Moles determination (e.g. measuring molar gas volume using a gas syringe, determining the formula of a hydrated salt by heating to constant mass)
- **PAG 2:** Titrations (acid-base and redox titrations, including back titrations)
- **PAG 3:** Enthalpy changes (e.g. calorimetry experiments measuring temperature changes for neutralisation, dissolution, or combustion)
- **PAG 4:** Rates of reaction (measuring how rate changes with concentration, temperature, or surface area using gas collection or colorimetry)
- **PAG 5:** Qualitative tests (identifying cations, anions, and functional groups using test-tube reactions)
- **PAG 6:** Organic preparations (synthesis, purification by recrystallisation, and testing of organic products such as aspirin or an ester)

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Exam Tip: Questions about practical skills appear across *all* papers, not just in a dedicated practical paper. You may be asked to plan experiments, analyse data, evaluate methods, or suggest improvements to procedures. Always link your answers to specific apparatus and techniques. The command words "describe," "explain," and "evaluate" carry different expectations — "describe" requires stating what you do, "explain" requires stating why, and "evaluate" requires assessing strengths and weaknesses with a supported conclusion.

1.6 Working with Data

Calculating Means

When calculating a mean, exclude any anomalous results first. An anomalous result is one that does not fit the pattern of the other data and falls well outside the expected range. You should identify and explain why a result might be anomalous (e.g. misreading a burette, contamination, incomplete reaction, air bubble in syringe). Report the mean to the appropriate number of significant figures or decimal places.

Percentage Yield and Atom Economy

$$\text{Percentage yield} = (\text{actual yield} \,/\, \text{theoretical yield}) \times 100\%$$

$$\text{Atom economy} = (\text{M}_\text{r} \text{ of desired product} \,/\, \text{sum of M}_\text{r} \text{ of all products}) \times 100\%$$

Percentage yield is always less than 100% in practice due to:

- Incomplete reactions (equilibrium not fully to the right)
- Side reactions producing unwanted by-products
- Losses during transfer between containers
- Losses during purification (e.g. filtration, evaporation, recrystallisation)
- Impure reactants

Atom economy is a *theoretical* measure of how efficiently a reaction uses its atoms. A high atom economy means less waste, which is important in green chemistry and industrial processes. Addition reactions

always have 100% atom economy because all atoms end up in a single product. Substitution and elimination reactions typically have lower atom economy because they produce by-products.

Worked Example — Atom Economy:

For the preparation of ethanol by fermentation:
$C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$

M_r of desired product ($2 \times C_2H_5OH$) = 2 × 46.0 = 92.0

Sum of M_r of all products = 92.0 + (2 × 44.0) = 180.0

Atom economy = (92.0 / 180.0) × 100 = **51.1%**

Compare with the hydration of ethene: $CH_2{=}CH_2 + H_2O \rightarrow C_2H_5OH$

Atom economy = 46.0 / 46.0 × 100 = **100%**
The hydration route has much better atom economy, but fermentation uses renewable feedstock (sugar), which is a green chemistry advantage.

Green chemistry: An approach to chemistry that seeks to reduce waste, use safer solvents and reagents, improve atom economy, use renewable feedstocks, design energy-efficient processes, and minimise environmental impact throughout the lifecycle of a chemical product. The 12 principles of green chemistry guide sustainable chemical manufacturing.

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